Earth and Planetary Science Letters, 94 (1989) 151–159
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

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Stability of remanence and paleomagnetic studies of some chromite ores from Barramiya and Allawi occurrences, Eastern Desert, Egypt

E. Refai [1], Nadia A. Wassif [2] and A. Shoaib [2]

[1] *Department of Geophysics, Cairo University, Giza (Egypt)*
[2] *Department of Geophysics, National Research Centre, Giza (Egypt)*

Received July 28, 1988; revised version received May 3, 1989

Rock magnetic investigations were carried out on natural Precambrian chromite ore at two occurrences in the Eastern Desert, Egypt. A medium/high coercivity component of remanence of reversed polarity can be defined. The mean direction of the five sites studied is $D = 198°$, $I = -44°$, with $k = 55$ and $\alpha_{95} = 10°$. Ore microscopic and magnetic examinations indicate the absence of magnetite and it is shown that the chromite mineral (ferrimagnetic phase) could be the carriers of the high stable NRM. The result of paleopole position of these Precambrian chromites are presented and discussed in the context of its age and the apparent polar wander path (APWP) of Africa.

1. Introduction

Previous studies on the Egyptian chromite ore were concerned mainly with the geological setting, field relations, chemical, mineralogical and X-ray investigations (e.g. [1–9] and many others) but no magnetic or paleomagnetic investigation of the ore have been published.

Samples were taken from two fresh exposures in Barramiya area (25°03′–25°10′N, 33°40′–34°05′E) and from three sites in Allawi area (24°30′–24°40′N, 33°45′–34°0′E). Oriented block samples of both chromite ores and the serpentinite host rock were collected; four to six samples at each site. The natural remanent magnetization (NRM) and the initial susceptibility were measured using an inductometer [10]. One to three cylindrical (2.5 cm diameter ×2.5 cm high) specimens were then cut from each sample and these specimens were subjected to progressive alternating field (AF) demagnetization to characterize the stability of magnetization. The magnetic mineralogy was investigated by reflected light microscopy and Curie temperature analysis.

2. Geological setting

The chromites investigated in this work are from the ultramafic rock mass of Barramiya and Allawi occurrences in the Eastern Desert which belongs to the Egyptian Precambrian basement complex (Fig. 1).

The stratigraphic classification of the basement complex in the Eastern Desert and Sinai was proposed by various workers, pioneered by Hume [11], Schurmann [12], followed by El-Ramly and Akaad [13], Akaad and Nowier [14,15] and El-Ramly [16]. Recently, El-Gaby [17] classified these Precambrian rocks into three groups: Hammamat

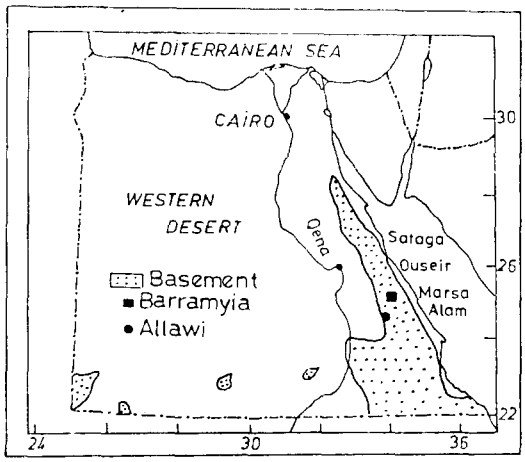


Fig. 1. Map of Egypt showing the outcrop of basement rocks (after El-Ramly and Akaad [13]). The approximate location of chromite studied is indicated.

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TABLE 1

Stratigraphic succession of the Egyptian basement complex [17]

Hammamat Group
Molasse-type clastics and penecontemporaneous calc-alkaline Dokhan volcanics, subvolcanic porphyry felsites, and plutonic calc-alkaline granites, including Old Gray granite and calc-alkaline younger granites.
Abu Ziran Group:
Metamorphosed andesite-rhyolite flows, tuffs and volcanoclastics (?)
Metasediments
Tholeiitic metavolcanics
Metagabbros
Serpentinized peridotites enclosing chromite lenses
Meatiq Group
Old continental basement (including Shaitian granite)
The rocks were diaphtorized at shallow depths and mobilized at greater depths

Group (youngest), Abu Ziran Group, and Meatiq Group (oldest). The rock sequence is given in Table 1.

The Egyptian basement complex lies within the foreland fold and thrust belt of the Pan-African Orogen. It reveals a series of swells along two geanticlines trending nearly NW–SE. The orogenic belt is dissected by two right-lateral shear zones, trending NEN–WSW, complementary to the left-lateral Najd fault system in the Arabian Peninsula.

The studied chromite ore bodies are lenses of variable sizes and shapes, mostly as sausage-like structures located along shear zones sharply separated from the country rocks and generally aligned NE–SW. The ore may also occur as small veins. Both lenses and veins are enclosed in serpentinites and talc carbonates. The chemical composition, mineralogy and origin of the numerous chromite deposits have been discussed by many authors ([6–9] and others), who concluded that the mineral paragenesis is identical that of Alpine ultramafic bodies. It is assumed that the dunites and peridotites, with their ore bodies were subjected to shearing movements during the folding of the area. It is possible that owing to the tectonic movement and local heating, the ore material becomes partly mobilized. Such partial mobilization will account for the sausage-like structures to thin veinlets and the strings connecting the lenses, the sharp contacts and the facts that the chromite lenses may occupy more than one horizon above the intrusion base. The temperature was probably high enough that an appreciable part of the chromite alteration took place during the mobil phase [6]. The mobil phase was followed by a third period of pneumatolitic hydrothermal phase in which the peridotites and dunites were transformed into serpentinites and talc carbonates.

3. Experimental results

3.1. Magnetic mineralogy and mineral chemistry

The mineralogy of the chromite ore and their related rocks has been determined by optical and chemical determinations together with some physical properties [18]. Magnetic intensity (σ_R) and susceptibility measurements (χ) of 11 samples from Barramiya and 15 samples from Allawi have indicated a considerable variation. The range values are as follows:

	σ_R ($\times 10^{-5}$ G)	χ ($\times 10^{-5}$ G/Oe)
Barramiya	45.76–170.30	41.30–135.50
Allawi	1.78– 14.040	0.32– 5.74

(1 G = 10^3 A/m, 1 G/Oe = 4π SI units (m/m^3)).

Chemical analysis for the major oxides of ten samples show also a considerable variation. On comparing the formulas of four representative samples, it is clear that the Barramiya chromite is generally poorer in aluminium and magnesium and richer in iron than the Allawi chromite.

Barramiya

1B: $Mg_{0.80}Fe_{0.62}Cr_{1.22}Al_{0.42}O_4$

3B: $Mg_{0.88}Fe_{0.41}Cr_{1.30}Al_{0.44}O_4$

Allawi

4L: $Mg_{0.92}Fe_{0.32}Cr_{0.91}Al_{0.89}O_4$

8L: $Mg_{0.82}Fe_{0.37}Cr_{1.04}Al_{0.82}O_4$

In reflected light, the color and reflectivity of polished surface varied considerably with composition. Samples from Allawi exhibit a characteristic greyish color and low reflectivity (normal type), while Barramiya samples were altered into a homogeneous single phase which has an intermediate color between magnetite and chromite. This phase appears pale grey in color with faint creamy to brownish tint and a higher reflectivity

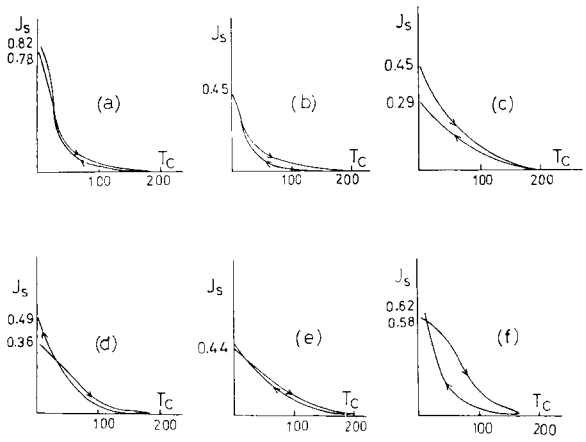


Fig. 2. Representative J_S-T curves for (a, b, c) Barramiya and (d, e, f) Allawi occurrences. Both heating and cooling cycles are shown and indicated by arrows.

than pure chromite but lower than that of magnetite. This phase is found to be highly magnetic with a magnetic intensity more than ten times higher and initial magnetic susceptibility values which are some twenty times higher than those of the Allawi chromite.

3.2. Thermomagnetic measurements

The saturation magnetization (J_S) as a function of temperature (T) was measured for ten samples (5 from each occurrence) using a horizontal automatic magnetic balance. Small chips from the measured chromite samples were ground to a coarse powder and heated in air in applied magnetic fields of about 310 mT. The thermomagnetic curves of most samples were fairly uniform in appearance and were nearly reversible and of the types illustrated in Fig. 2a–e. Only one (Fig. 2f) is clearly irreversible. Some samples showed a more rapid initial decrease in magnetization which can be attributed to the presence of an appreciable fraction of paramagnetic mineral. The characteristic Curie temperature is about 90 °C for samples of Barramiya and about 135 °C for Allawi samples. These Curie temperatures indicate that the ferrimagnetic phase present is very much different from that of magnetite ($T_c = 580$ °C). In addition to the major magnetic phase, there seems to be a second minor one which permit the smooth decay of the curves up to a temperature of about 200 °C for all samples. The saturation magnetization of the samples is low (see Fig. 2).

3.3. Alternating field (AF) demagnetization

The magnetic measurements have been done with a Schonstedt computer-assisted spinner magnetometer. Demagnetization procedures have been carried out with Schonstedt equipment (alternating field demagnetizing apparatus GSD-1 in mumetal shields with a very low residual field of about 10 nT).

The specimens were treated by AF increased in 15–20 steps up to 100 mT. The directions of both soft and hard component can be analyzed from the Zijderveld [19] vector diagram. On the basis of this study the results can be divided into two categories.

Stable sites (Allawi sites). One of the pilot specimens (5L – 1) is unaffected by AF demagnetization (Fig. 3). The end points gave rise to consistent, but small changes in both intensity and direction so that it is difficult to differentiate between the secondary and the characteristic re-

TABLE 2

The mean NRM directions and paleopole positions for the sites studied and their statistical parameters

Sites	N	D_m (°)	I_m (°)	k	α_{95} (°)	ODF (mT)	VGP	
							°N	°E
1B	6	202	−51	42	10	27.5	−69.5	279.4
2B	5	195	−55	14	21	25.0	−73.4	261.6
3L	5	210	−39	21	17	20.0	−62.3	302.5
4L	4	188	−39	35	16	27.5	−82.0	285.2
5L	6	196	−35	45	10	25.0	−74.2	287.3
Mean	5	198	−44	55	10		−73.5	292.9
							$K = 80$; $A_{95} = 8.6$°;	

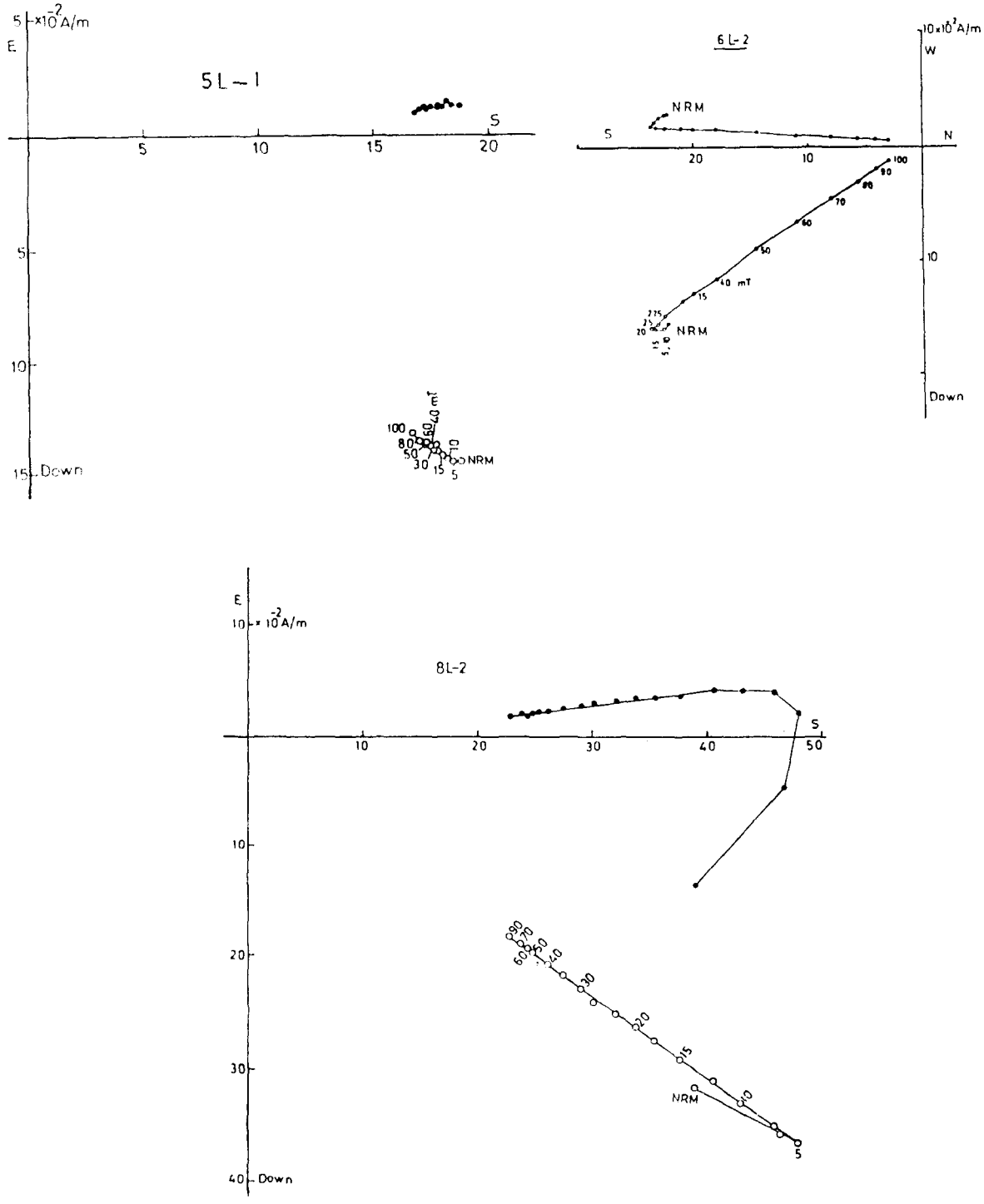


Fig. 3. Vector diagram showing the variation of remanence vector during progressive AF demagnetization. Numerical figures besides circles indicate the strength of demagnetizing field intensity along the axes is in SI units. Solid circles represent projections on the horizontal plane, and open circles those on the vertical plane.

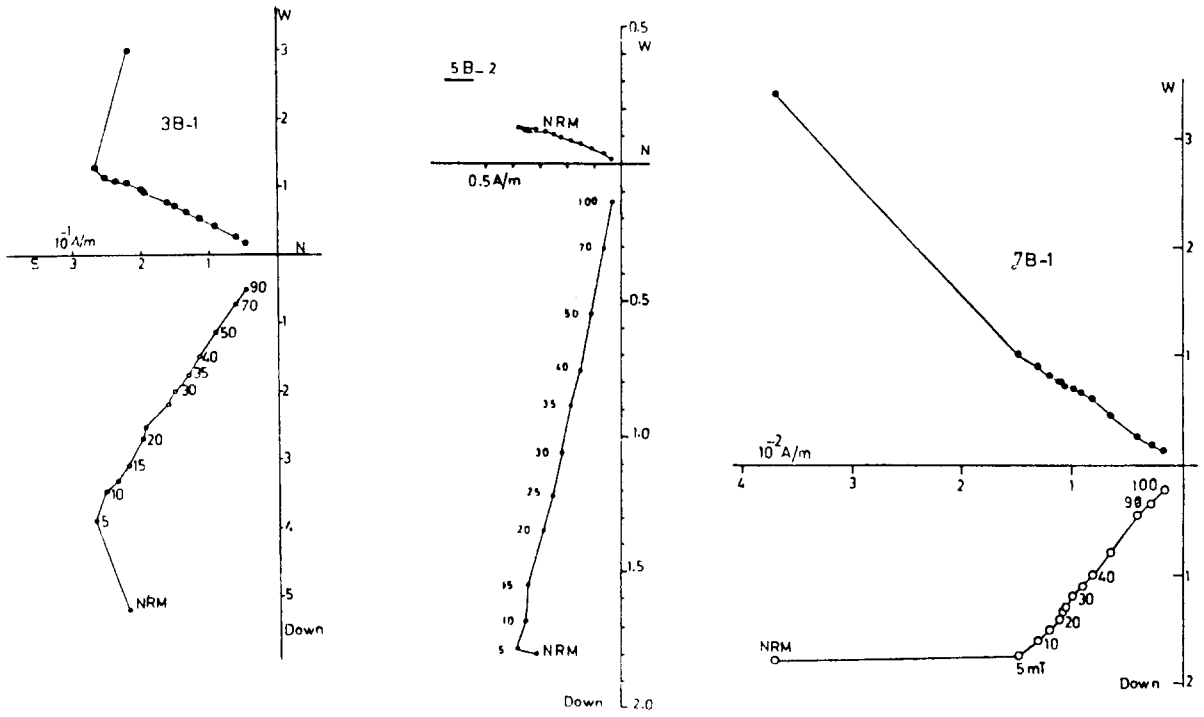


Fig. 3. (continued).

manent magnetization. Another two examples (Fig. 3, specimens 6L-2 and 8L-2) explicate the variations in the NRM intensity on the orthogonal projection, whereas the specimens yield a single characteristic magnetization in which an unstable component shows generally an intermediate downward inclination. These specimens (6L-2 and 8L-2) have typical "knee-shaped" intensity decay curves and high median destructive fields, 60 mT and more (Fig. 4).

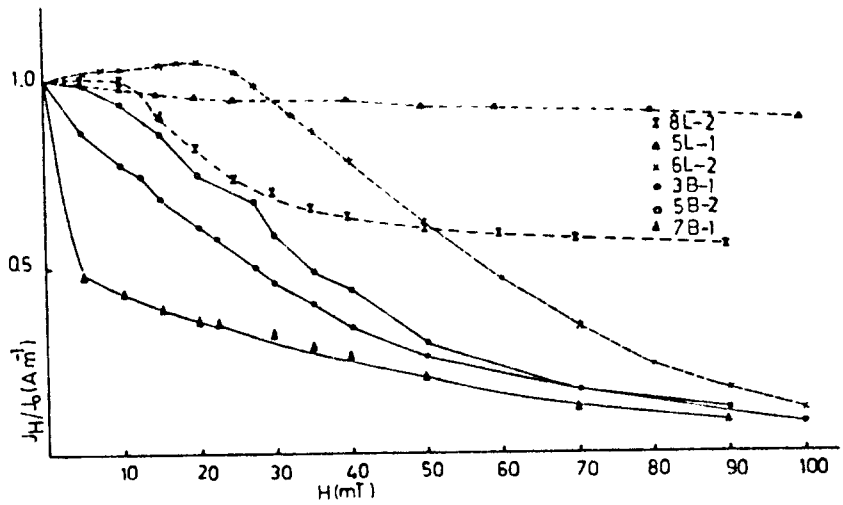


Fig. 4. Normalized AF intensity decay curves for specimens from Allawi (dashed lines) and Barramiya (solid lines).

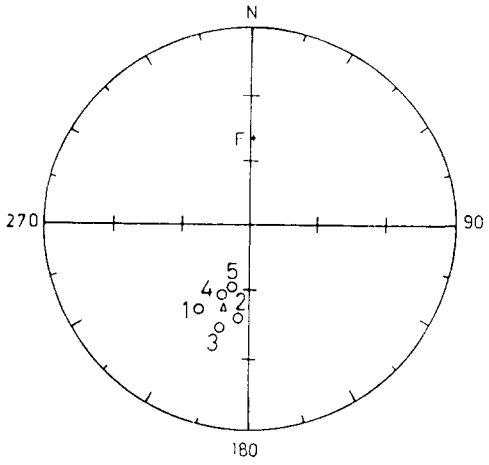


Fig. 5. Stereoplot showing the mean directions of the five sites. F = present field direction, Δ = the mean direction for all sites. Site numbers are given.

Intermediate sites (Barramiya sites). The pilot specimens (3B-1, 5B-2 and 7B-1) from the Barramiya sites show slowly decreasing intensity decay curves (Fig. 4) and median destructive fields of 27.5 mT, 35 mT and 5 mT, respectively. AF demagnetization (Fig. 3) showed that a field of about 25 mT was sufficient to reduce half the value of the magnetic moment. The direction of the unstable soft component is steeply downward.

3.4. Paleomagnetic data

The primary NRM direction was taken to be the stable end point reached by the NRM vector during AF demagnetization, or the direction given by high AF linear segments in the Zijderveld diagram. Table 2 lists the mean NRM direction

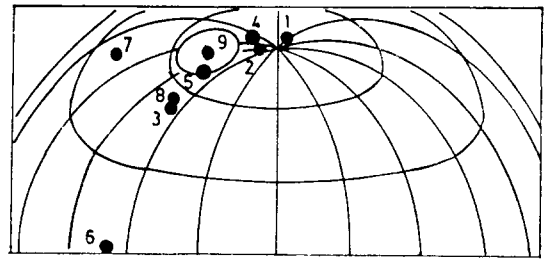


Fig. 6. Lambert equal area projection of the pole positions from Table 3.

for each site and the associated statistical parameters. These parameters were calculated giving unit weight to each sample [20]. The direction of magnetization of the samples from all five sites was reversed, i.e. the inclination was always negative and the declinations were relatively consistent to the southwest. The results are fairly well concentrated (Fig. 5) and have a mean direction for the five sites of $D_m = 198°$, $I_m = -44°$ ($k = 55$, $\alpha_{95} = 10°$)—without tectonic correction. A mean paleopole position at $-73.5°$ N, 292.9° and $A_{95} = 8.6°$ is calculated.

The time of consolidation of these chromites can probably be estimated by a comparison of the poles with the hitherto established pole positions which fall in close proximity. Davies et al. [21] gave some paleomagnetic data of certain late Precambrian and Early Paleozoic rocks from the Red Sea Hills, Eastern Desert Egypt; their results are given in Table 3. Saradeth et al. [22] made contributions to the African polar wander path in southwest Egypt and north Sudan. They show that the poles of Nabati and Bir Safsaf (Table 3) confirm a

TABLE 3

Late Precambrian to lower Paleozoic paleopole position from Egypt and Sudan

Pole	Region and rock unit	Age	Paleopole position		95 (°)	Reference
			lat.	long.		
1	Egypt, Quena-safaga dykes	480–530 Ma	87° N	304° E	5	Devies et al. [21]
2	Egypt, um Rus dykes	464–497 Ma	86° N	185° E	6	
3	Egypt, Dokhan volcanics	500 Ma	54° N	327° E	15	Devies et al. [21]
	Egypt, Dokhan volcanics	500–603 Ma	36° S	17° E	17	
4	Egypt, Bir safsaf (granodioritic porphyry dykes)	Cambrian (502 Ma)	80° N	249° E	9.2	Saradeth et al. [22]
5	Sudan, Nabati complex	580 Ma	67.9° N	314.1° E	13.6	Saradeth et al. [22]
6	Sudan, Kadaweb (gabbroic unit)	720 Ma	1.0° N	319.6° E	19.8	Saradeth et al. [22]
7	Egypt, El Shadli geosynclinal	Upper Proterozoic	47° N	256° E	18.9	Shereef et al. [22]
8	Egypt, El shadli geosynclinal	Ordivician–Cambrian	−57° N	324° E	25	
9	Egypt, chromite	Late Precambrian	−73.5° N	292.9° E	10	this work

lower Paleozoic pole position to the north of Africa (Fig. 6). Also the pole positions of the geosynclinal metavolcanics obtained recently by Shereef et al. [23] is shown in Fig. 6. Finally, the pole of these chromites is also given as a further contribution to fill the data gap for African poles in the Late Precambrian period.

Generally, authors agree recently that the Abu Ziran and Hammamat Groups (Table 1) comprise the rock associations developed during the Pan-African Cycle, ca. 1000–500 Ma [24]. The Meatiq Group represents the stratigraphically oldest rock unit over which the Pan-African Abu Ziran Group has been accumulated or thrust. The age of the Meatiq Group must be at least 1770 Ma which is the U-Pb zircon age obtained from the foliated granite of Wadi Abu Rusheid–Wadi Sikait [25]. Available isotopic ages reported from Pan-African granites from Egypt [26,27] fall well within the time span 670–550 Ma. Dokhan volcanics have Rb-Sr ages of 665-654 Ma [26], 618–602 Ma [28] and 616 Ma (Ries and Derbyshire, in [29]). Accordingly, the probable age for the Egyptian serpentinites (including the chromite lenses, Table 1) must be older than 670 Ma.

4. Discussion and conclusion

It is clear that the remanent magnetism residing in the chromite ore is fairly stable and is SSW and moderately steeply downward (Table 2, Fig. 5). The normalized curves for these specimens (Fig. 4) suggest that stable magnetization is of thermoremanent or thermochemical origin. It is generally accepted that chemical remanent magnetization (CRM) is acquired when magnetic substance forms in a magnetic field, either by chemical changes in a pre-existing solid substance or by crystallization from a solution (or from a melt) if this occurs below the Curie point. The high-iron chromites from Nausahi, India, resemble magnetite in physical properties and in being moderately to strongly magnetic (see [30]). It is not possible from the (AF) decay curves to distinguish between these two types of magnetization (CRM or TRM). On the basis of the demagnetization experiment, it is evident that the median destructive field ($H_{0.5}$) varies considerably—the Barramiya specimens show relatively low $H_{0.5}$ values (5–35 mT), while the Allawi specimens show extremely stable directions with values greater than 60 mT. However, such very high $H_{0.5}$ fields of these chromite spinels obtained during AF demagnetization are very similar to those predicted theoretically for single domain or pseudo-single domain grains of almost pure magnetite [31]. Kumar and Bhalla [32,33] mentioned that such high fields indicate that chromites have excellent time stability and hence are efficient carriers of remanence over long periods.

Zoned chromite (formed of two or more zones of ferrian chomite and chromian magnetite) has been recorded in serpentinite-peridotite bodies and their associated chromite cumulates particularly those belonging to the Alpine-type ultramafics [6,8,9,34,35]. Ore microscopic studies of the present chromite and their host rocks are free from such alteration zoned grains and also free from primary or secondary magnetite. Since magnetite is by far the most abundant magnetic product of serpentinization, the alteration of chromite is generally connected with the degree of serpentinization and consequently with the amount of magnetite released.

Kumar et al. [32,33], in their study of Sukinda chromite ore, mentioned that "if the alteration during serpentinization was sufficient to produce enough magnetite; the primary direction of magnetization should be overprinted by a secondary one representing the time of serpentinization". Inspections of the AF demagnetization results proved the absence of such magnetization. These observations mean that the degree of alteration was not enough to form a considerable amount of magnetite. Thus it may be suggested that the stable component of magnetization (specially Allawi samples) was acquired during the primary consolidation of the chromite. But, possibly, for Barramiya samples (which chemically show a remarkably high iron content and consequently high values of σ_R and χ) it can be said that the serpentinization processes are concerned mainly with the enrichment of the ore by iron and its transformation to highly magnetic homogeneous phase. During thermal treatment, the Curie temperature of the studied chromites were estimated to range from about 90° to 135°C. However, the J_S-T synthesized data [35–37] do not indicate a definite Curie temperature.

Finally, on the basis of the rock magnetic and paleomagnetic results, the following conclusions seem reasonable:

(1) A directionally stable component of NRM was isolated by the AF demagnetization in individual samples of the two occurrences. Median destructive field were very high for Allawi chromites (760 mT) but relatively low (5–35.0 mT) for the Barramiya group, although sufficient to ensure long-term stability of the measured NRM.

(2) With the absence of magnetic oxides (magnetite and hematite), the magnetic mineral responsible for carrying such highly stable remanence of the ore, is most probably the ferrimagnetic phase of the chromite itself, i.e. the chromites are themselves capable of carrying hard and stable remanence similar to single-domain iron oxides.

(3) Despite the presence of the chromite lenses in serpentinites and related rocks, especially for Allawi samples, the processes of serpentinization has not produced magnetite in the ore (as seen from the microscopic and thermal examinations). But, probably for Barramiya samples there is some evidence for the transformation of the ore to a highly magnetic homogeneous phase.

(4) Kumar et al. [32,33], in their study of Sukinda chromite, mentioned that the consistent paleomagnetic directions of remanent magnetism suggested that these chromite deposits were of a stratiform type. However, in this present study, the chromite with host serpentinites rocks are classified as Alpine type. The stable reversed directions obtained and the consistent paleomagnetic results suggest that the chromite lenses have been consolidated and arranged in an accurate pattern in a fairly stable tectonic conditions possibly later than the partial mobile phase.

(5) Paleomagnetic data point to a Late Precambrian age for these chromites. The geological sequence of the basement complex and available isotope ages for younger components indicate that the chromite studied may be roughly older than 670 Ma.

Acknowledgements

The authors wish to thank Prof. D.H. Tarling for critically reading the manuscript and Prof. Dr. M.A. Takala (Cairo university) for his help and valuable discussions during mineralogical work. All the magnetic cleaning were done at the Geological Survey and Mining Authority. The efforts by Mr. Helmy Abd El-Rasik and Mr. Rifaat A. Nashid are gratefully acknowledged.

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